

Surface waves and the shallow-water limit

Physics for a hands-on numerical laboratory

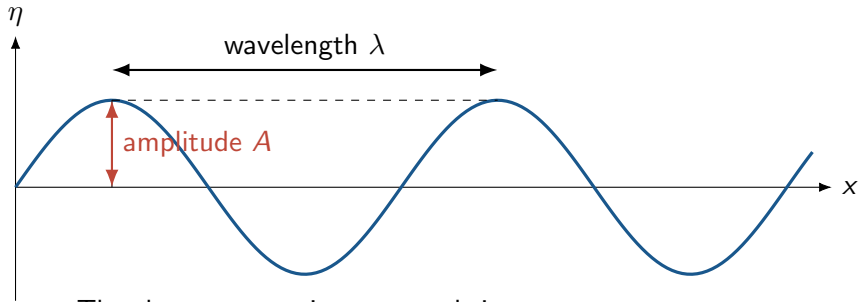
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September 28, 2026

What should you be able to do?

- Read a progressive-wave expression and infer its direction and speed.
- Decide whether a surface wave is in the shallow-water regime.
- Predict the long-wave speed $c = \sqrt{gH}$ and a travel time.
- Explain reflection and the natural period of a closed basin.

Wave vocabulary



The shape repeats in space and time.

Period T ; frequency $f_w = 1/T$; angular frequency $\omega = 2\pi/T$; wavenumber $k = 2\pi/\lambda$.

A progressive sinusoidal wave

$$\eta(x, t) = A \cos(kx - \omega t + \phi_0).$$

A fixed phase satisfies

$$kx - \omega t = \text{constant} \quad \Rightarrow \quad \frac{dx}{dt} = \frac{\omega}{k} = c_p.$$

- $kx - \omega t$: propagation toward increasing x .
- $kx + \omega t$: propagation toward decreasing x .
- $c_p = \omega/k = \lambda/T$ is the phase speed.

A crest is a particular phase, not a marked parcel of water.

Exercise 1: read the wave

Consider, with x in metres and t in seconds,

$$\eta(x, t) = 0.15 \cos \left(\frac{2\pi x}{200} - \frac{2\pi t}{20} \right) \text{ m.}$$

Find:

- 1 amplitude, wavelength, and period;
- 2 propagation direction and phase speed;
- 3 η at $x = 50 \text{ m}$ and $t = 2 \text{ s}$.

Work with a neighbour for 4 minutes. Keep units in every answer.

Exercise 1: solution

$$A = 0.15 \text{ m},$$

$$\lambda = 200 \text{ m},$$

$$T = 20 \text{ s},$$

$$c_p = \frac{\lambda}{T} = 10 \text{ m s}^{-1}.$$

The minus sign gives propagation toward increasing x . At $x = 50 \text{ m}$ and $t = 2 \text{ s}$,

$$kx - \omega t = \frac{\pi}{2} - \frac{\pi}{5} = \frac{3\pi}{10}, \quad \eta \approx 0.088 \text{ m}.$$

Check

A surface displacement cannot exceed the stated amplitude 0.15 m .

Surface gravity is the restoring force

A displaced free surface creates a horizontal pressure gradient. Gravity accelerates water away from high surface elevation.

$$\frac{\partial u}{\partial t} = -g \frac{\partial \eta}{\partial x} \quad (\text{simplest linear, non-rotating form}).$$

Two geometric scales matter

Water depth H and wavelength λ determine whether the bottom affects the entire water-column motion.

The surface-gravity-wave dispersion relation

For linear waves over uniform depth,

$$\omega^2 = gk \tanh(kH).$$

- It connects frequency, wavelength, gravity, and water depth.
- The dimensionless parameter $kH = 2\pi H/\lambda$ identifies the regime.
- If ω/k depends on k , different wavelengths propagate at different phase speeds: the waves are dispersive.

Deep and shallow limits

Deep water: $kH \gg 1$

$$\tanh(kH) \approx 1$$

$$\omega^2 \approx gk, \quad c_p \approx \sqrt{\frac{g}{k}}.$$

Speed depends on wavelength.

Shallow water: $kH \ll 1$

$$\tanh(kH) \approx kH$$

$$\omega^2 \approx gHk^2, \quad c_p \approx \sqrt{gH}.$$

Speed is independent of wavelength.

“Shallow” compares depth with wavelength, not with a swimmer or ship.

Exercise 2: classify the waves

Estimate $kH = 2\pi H/\lambda$ and classify each case.

Example	H	λ
Long ocean disturbance	4000 m	200 km
Wind wave on a shelf	50 m	100 m
Basin-scale tide	100 m	1000 km

Which examples can a non-dispersive shallow-water model represent most directly?

Exercise 2: solution

Example	kH	Interpretation
Long ocean disturbance	0.126	shallow-water approximation
Wind wave on a shelf	3.14	not shallow; dispersive
Basin-scale tide	0.00063	strongly shallow

Important

A tsunami and a tide can both satisfy shallow-water dynamics even though one begins as an initial disturbance and the other is continuously forced.

What varying depth changes

If $H = H(x, y)$, the local long-wave speed is

$$c(x, y) = \sqrt{gH(x, y)}.$$

- A wave slows as it enters shallower water.
- At approximately conserved frequency, its wavelength shortens.
- Amplitude can change through shoaling, convergence/divergence, and reflection.
- An abrupt depth change usually reflects more strongly than a gentle change.

Exercise 3: depth and travel time

Compare a long wave over depths $H_1 = 100$ m and $H_2 = 400$ m.

- 1 Calculate both speeds.
- 2 Find the speed ratio c_2/c_1 before using a calculator.
- 3 Estimate the travel time across 1000 km in each case.

Use $g = 9.81 \text{ m s}^{-2}$.

Exercise 3: solution

$$c_1 = \sqrt{9.81 \times 100} = 31.3 \text{ m s}^{-1},$$

$$c_2 = \sqrt{9.81 \times 400} = 62.6 \text{ m s}^{-1},$$

$$\frac{c_2}{c_1} = \sqrt{\frac{400}{100}} = 2.$$

For $L = 1000 \text{ km}$, $t = L/c$ gives approximately

$$t_1 = 8.9 \text{ h}, \quad t_2 = 4.4 \text{ h}.$$

Deeper water gives faster long waves.

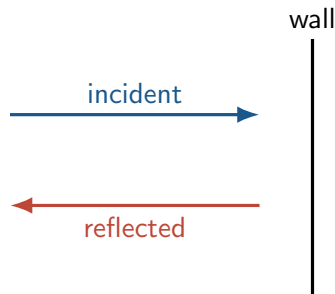
Reflection from a vertical coast

At a solid wall, normal velocity must vanish:

$$u_n = 0.$$

An incident and reflected wave combine to satisfy this condition.

- Normal velocity reverses sign.
- Surface displacement reflects without a sign reversal at a rigid wall.
- Incident and reflected waves may form a standing pattern.



Standing waves and basin seiches

Two equal waves moving in opposite directions give

$$\begin{aligned}\eta &= A \cos(kx - \omega t) + A \cos(kx + \omega t) \\ &= 2A \cos(kx) \cos(\omega t).\end{aligned}$$

The spatial pattern has fixed nodes and antinodes. For the fundamental mode of a closed rectangular basin,

$$\lambda_1 = 2L, \quad T_1 = \frac{2L}{\sqrt{gH}}.$$

Repeated forcing near a natural period can produce a large response.

Exercise 4: a basin timescale

A closed basin has length $L = 600$ km and uniform depth $H = 100$ m.

- 1 Estimate its fundamental seiche period.
- 2 If the basin length doubles at the same depth, what happens to the period?
- 3 If depth quadruples at the same length, what happens to the period?

Exercise 4: solution

$$c = 31.3 \text{ m s}^{-1}, \quad T_1 = \frac{2(600 \times 10^3)}{31.3} \approx 10.6 \text{ h}.$$

- Doubling L doubles T_1 .
- Quadrupling H doubles c and therefore halves T_1 .

These are hypotheses that can be tested directly in Part A or Part C.